WORKED SOLUTION

STEP II 1991, Question 10

Pure • Spherical polar coordinates

STEP II 1991, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q10

The key idea

Despite the database tag, this is a question about lines and planes in three dimensions. The tip of the shadow is where the ray of sunlight that grazes the top of the stick meets the ground. So we write the ray as a line through the Sun and the tip of the stick, write the ground as a plane through the base perpendicular to the stick, and intersect them. Everything is easy once we write all positions in terms of the unit vector along the stick.

Solution

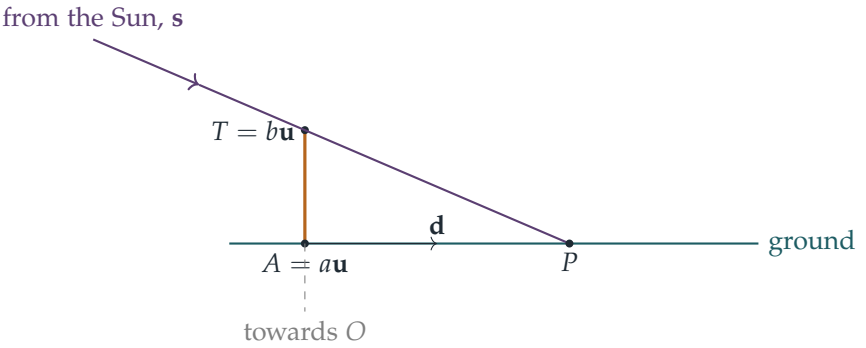
Let O be the centre of the Earth, and write

$$\mathbf{u} = \begin{pmatrix} 0 \\ \cos\phi \\ \sin\phi \end{pmatrix}, \qquad \mathbf{s} = R\begin{pmatrix} \cos\theta \\ \sin\theta \\ 0 \end{pmatrix}.$$

Then $\mathbf{u}$ is a unit vector along the stick, the base is $A = a\mathbf{u}$, the tip is $T = b\mathbf{u}$, and the Sun is at $\mathbf{s}$. The one dot product we need is

$$\mathbf{s} \cdot \mathbf{u} = R\sin\theta\cos\phi.$$

We write $c = R\cos\phi\sin\theta$ for short.



The ray through the tip

Light reaching the tip of the shadow, P, has just passed the tip of the stick. So P lies on the line through $\mathbf{s}$ and $b\mathbf{u}$:

$$\mathbf{p} = \mathbf{s} + t\,(b\mathbf{u} - \mathbf{s}) = (1-t)\,\mathbf{s} + tb\,\mathbf{u}$$

for some real t.

The ground

The ground is horizontal, meaning perpendicular to the stick, and it passes through the base $a\mathbf{u}$. Since $\mathbf{u}\cdot\mathbf{u} = 1$, it is the plane $\mathbf{r}\cdot\mathbf{u} = a$.

Intersecting

Substituting the line into the plane, and using $\mathbf{u}\cdot\mathbf{u} = 1$ and $\mathbf{s}\cdot\mathbf{u} = c$,

$$(1-t)c + tb = a \qquad \Longrightarrow \qquad t = \frac{c-a}{c-b}.$$

We can divide by $c-b$ because the Sun is shining on the stick. The Sun is above the horizon at the tip exactly when it is on the far side of the tangent plane there, i.e. $\mathbf{s}\cdot\mathbf{u} > b$, so $c > b$.

> **Ask yourself** Is this P really on the ground beyond the stick, and not somewhere behind the Sun?

Since $c > b > a$, we have $c-a > c-b > 0$, so $t > 1$. The point $t = 0$ is the Sun and $t = 1$ is the tip of the stick, so $t > 1$ is past the tip, travelling away from the Sun, as it should be.

From the value of t,

$$1 - t = \frac{(c-b)-(c-a)}{c-b} = \frac{a-b}{c-b} = -\frac{h}{c-b},$$
$$tb - a = \frac{b(c-a)-a(c-b)}{c-b} = \frac{c(b-a)}{c-b} = \frac{ch}{c-b}.$$

So the displacement from the base to the tip of the shadow is

$$\begin{aligned}
\mathbf{d} = \mathbf{p} - a\mathbf{u} &= (1-t)\,\mathbf{s} + (tb-a)\,\mathbf{u} \\
&= \frac{h}{c-b}(c\,\mathbf{u} - \mathbf{s}) \\
&= \frac{Rh}{c-b}\left[\cos\phi\sin\theta\begin{pmatrix}0\\ \cos\phi\\ \sin\phi\end{pmatrix} - \begin{pmatrix}\cos\theta\\ \sin\theta\\ 0\end{pmatrix}\right] \\
&= \frac{Rh}{c-b}\begin{pmatrix}-\cos\theta\\ -(1-\cos^2\phi)\sin\theta\\ \cos\phi\sin\phi\sin\theta\end{pmatrix}.
\end{aligned}$$

Since $1-\cos^2\phi = \sin^2\phi$ and $c - b = R\cos\phi\sin\theta - b$, this is

$$\mathbf{d} = Rh\,(R\cos\phi\sin\theta - b)^{-1}\begin{pmatrix} -\cos\theta \\ -\sin^2\phi\sin\theta \\ \cos\phi\sin\phi\sin\theta \end{pmatrix},$$

as required.

The middle line, $\mathbf{d} = \frac{h}{c-b}(c\,\mathbf{u} - \mathbf{s})$, is worth noticing. The vector $c\,\mathbf{u}$ is the component of $\mathbf{s}$ along the stick, so $c\,\mathbf{u} - \mathbf{s}$ is minus the horizontal part of the Sun's position vector. The shadow points directly away from the Sun's horizontal direction, as experience says it should.

Checks

- $\mathbf{d}\cdot\mathbf{u} = \frac{Rh}{c-b}\left(-\sin^2\phi\sin\theta\cos\phi + \cos\phi\sin^2\phi\sin\theta\right) = 0$, so the shadow lies along the ground.
- Take $\phi = 0$, so the stick points along the y-axis. Then $\mathbf{d} = \frac{Rh}{R\sin\theta - b}(-\cos\theta, 0, 0)$. When $\theta = \frac{\pi}{2}$ the Sun is directly overhead and the shadow vanishes. When R is huge, the length tends to $h\cot\theta$. That is the familiar flat-Earth answer for a Sun at elevation θ.
- Numerically, we located the shadow tip directly, by intersecting the ray with the ground plane, for thousands of random choices of R, a, h, θ, ϕ with the Sun above the horizon. It always agreed with the formula, with a largest relative error of about 10^{-13}.

Common slips

- Taking the ground to be the plane $y = a\cos\phi$, or some other coordinate plane. "Horizontal" means perpendicular to $\mathbf{u}$, which is $\mathbf{r}\cdot\mathbf{u} = a$.
- Expanding everything in components from the start. Keeping $\mathbf{u}$ and $\mathbf{s}$ as vectors until the last line saves a lot of algebra.

Beyond STEP

Here sunlight is a family of lines through one point, the Sun, so the shadow is a *central projection* of the stick onto the ground plane. When the Sun is very far away, the rays are almost parallel and the projection becomes a parallel projection. That is the kind of map linear algebra studies, and the vector $c\,\mathbf{u} - \mathbf{s}$ is the projection of $\mathbf{s}$ onto the stick minus $\mathbf{s}$ itself. Strang's *Introduction to Linear Algebra* treats projections onto lines and planes in detail.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.